PRELIMINARY DESIGN – FOR MDR INTEGRATION

Raspberry Pi Pico Electronics Architecture

Process flow, component rationale and 3K design-criteria fit

RTC 2027 Student 3K | Mission Design Report: electronics contribution | 4 November 2026

Scope and status. This is the electronics team's contribution draft to the team's single MDR; it is not a separately submitted report and not a completed flight design. Competition constraints are attributed to the organiser documents [1], [2], [3]. Thresholds, rates, pin allocations, power figures and the thermal payload are *team proposals* marked as such, and require simulation, approval and later testing in line with the MDR/PDR/CDR maturity expectations [3], [4]. Manufacturer figures are quoted from the cited datasheets and should be re-checked against the exact module purchased.

1. Design drivers: how the Pico design meets the 3K criteria

The design uses **three separate Raspberry Pi Pico boards** (RP2040) [5], [6]: one *logger/payload* Pico that never commands recovery, and two *independent recovery* Picos (primary and backup), each with its own barometer and battery. This keeps required flight-data logging and the thermal experiment from sharing a failure path with parachute deployment, while using one familiar, low-cost, well-documented platform for every computer in the rocket.

3K criterion	Design response	Basis / status
Record altitude, velocity, acceleration, GPS trajectory	Logger Pico records BMP280 pressure-altitude and MPU6050 acceleration with timestamps; velocity is derived by filtering/fusion; GPS track is taken from the organiser beacon as a separate dataset.	[1] Sec. 2.1; proposed method
Detect launch, apogee, descent	Multi-condition state machine (Figs. 2–3) with lock-outs and time-out back-ups.	[1] Sec. 4.2.2.4; proposed
Two-stage recovery: drogue at apogee, main at lower altitude	Two recovery Picos each execute both events; main altitude selected in the 150–450 m window, below the 500 m inspection criterion.	[1] 4.2.2.4; [7] Row 8
Independent / redundant activation, release redundancy	Primary and backup recovery Picos, separate batteries, separate barometers, outputs held low in hardware (Fig. 5).	[1] 4.1.3, 4.2.2.2; mechanism TBD
Safe/armed states, arming confirmation	Arming switch gates power; outputs forced low by pull-downs; launch-inhibit logic; buzzer/LED status code.	[1] 4.6.3–4.6.7
Vibration, impact, sensor error	Plausibility gates, filtering, CRC records, strain-relieved connectors, watchdog warm restart.	[1] 4.1.2, 4.4
$>$ 1.5 h electrical operation	Single Li-ion cell per Pico; load ≈56 mA estimated, 200 mA design basis (Sec. 8).	[1] 4.7.2; [7] Row 6
Official GPS tracker	Separate, organiser-standard beacon, own power, no electrical dependence on Pico boards.	[1] 2.2, 5.1
Additional payload	Two DS18B20 probes read by the logger Pico: internal avionics-bay vs inner-wall temperature difference; retained in rocket, no radio, no stored-energy device.	[1] 7.1; team proposal
No unapproved radio emissions	Plain Pico (not Pico W): no on-board radio.	[1] 4.1.4; [5]

2. System architecture

Critical interface rule. The logger never commands recovery, and nothing on Domain L (SD card, thermal probes, buzzer) can block a recovery decision. Cross-links between domains are limited to a common ground reference at the release interface if the recovery team requires it; no shared supply [1].

3. Why the Raspberry Pi Pico

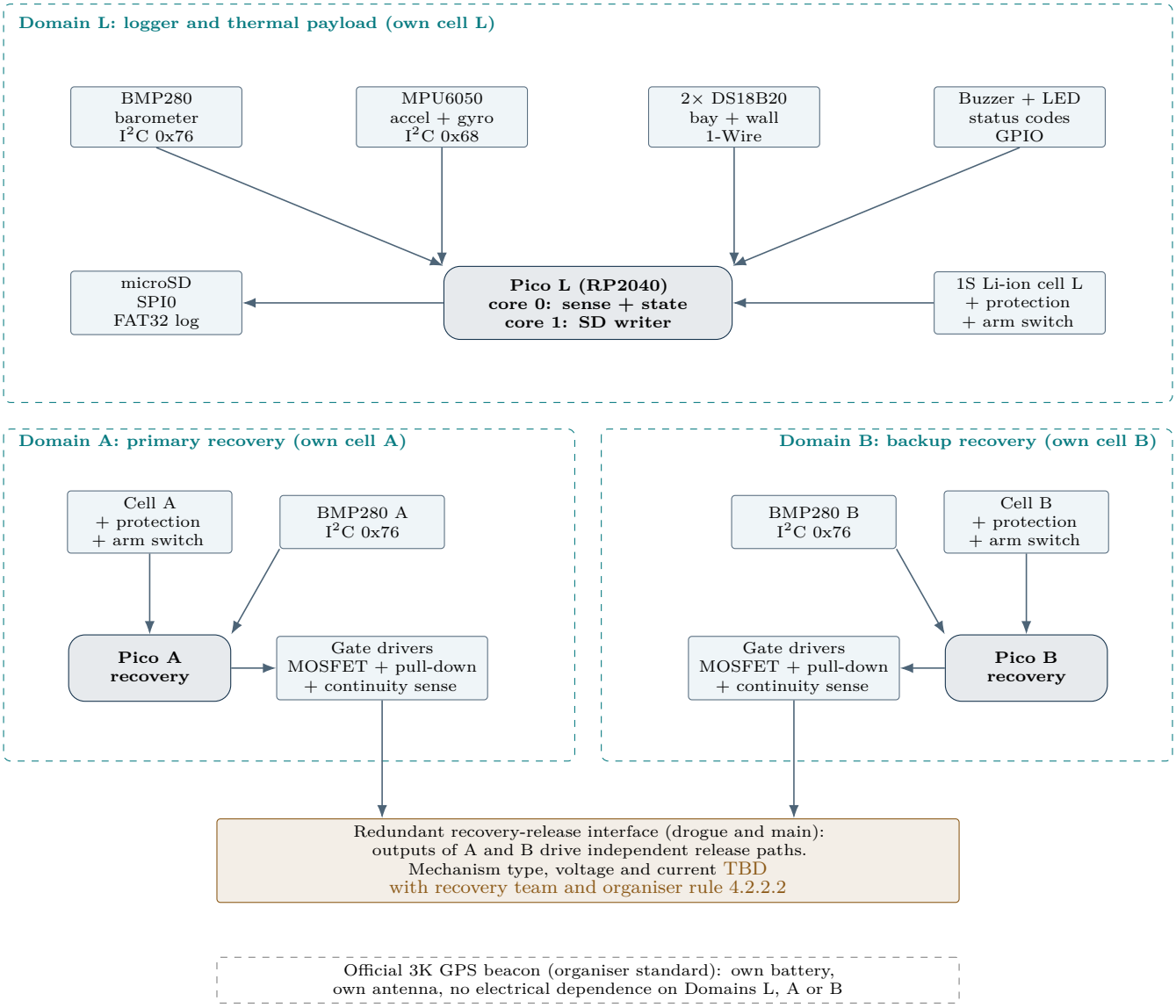


Figure 1: Electronics architecture. Three Pico boards in three power domains; the logger can fail without affecting recovery, and recovery A can fail without affecting recovery B. Illustrative; wiring and fault independence remain to be approved.

RP2040 / Pico feature	What it provides	Benefit to the 3K rocket
Dual Cortex-M0+ cores, up to 133 MHz	Two independent cores sharing memory [6].	Core 0 senses and decides; core 1 does slow SD writes, so storage latency cannot delay sampling (Fig. 4).
264 kB on-chip SRAM	Fast RAM for buffers [6].	A 5 s pre-trigger and write-ahead buffer at 24 B × 200 Hz uses about 24 kB (≈9% of SRAM).
Hardware I²C, SPI, UART, ADC	Two I²C, two SPI, two UART, 12-bit ADC inputs [6].	Covers IMU/baro (I²C), SD (SPI) and battery-voltage monitoring without extra chips.
Programmable I/O (PIO)	Eight state machines for custom protocols [6].	Cycle-exact 1-Wire for DS18B20 without blocking the CPU.
Watchdog with scratch registers	Reset timer; scratch registers survive a watchdog reset [6].	In-flight reset recovers flight state (Fig. 2): robustness against vibration/brown-out glitches.
Wide supply: VSYS 1.8–5.5 V	On-board buck-boost supply to 3.3 V [5].	A single 3.0–4.2 V Li-ion cell powers the board directly; the rail holds as the cell discharges.
No radio (Pico, not Pico W)	Plain board without Wi-Fi/Bluetooth [5].	Avoids unapproved emissions from the rocket [1].
Open C/C++ SDK, SWD debug, UF2 boot	Free toolchain and bootloader [8].	Rapid iteration, scripted builds, and the same toolchain on all three boards.

Feature	What it provides	Benefit (continued)
Low cost and availability	Retail boards at a few ringgit, drop-in spares.	Three flight computers plus spares fit a student budget; identical hardware simplifies sparing.

Alternatives considered. An 8-bit AVR such as the ATmega328P has only 2 kB of SRAM [9], too little for a buffered logger plus filtering. ESP32 devices offer more processing but include Wi-Fi/Bluetooth radios [10] that must be disabled or justified under the radio rules [1]. The Pico gives enough memory and I/O without a radio at similar cost.

Known Pico limits and mitigations. The Cortex-M0+ has no hardware FPU [6]; at 200 Hz the filter workload is small, but fixed-point or measured float timing will be checked. Flash is external (XIP), so *no flash writes during flight*; recovery events are kept in RAM and dumped after landing [6], [8]. A Pico dev board is not qualified flight hardware: the independence of A and B, the watchdog, hard-wired pull-downs and testing are what give confidence, not the board itself. The Pico operating range is −20 to +85 °C [5]; the bay temperature is measured by the payload.

4. Component-by-component rationale

Component	Role	Rationale and fit to 3K criteria	Limit / mitigation
Pico L (logger)	Flight data, state estimate, thermal payload, status.	Satisfies mandatory output list [1]; isolates non-safety functions from recovery.	Not a recovery path. Verified by Fig. 1 separation.
Pico A / Pico B (recovery)	Independent apogee and main-altitude logic and output control.	Redundant activation and release redundancy [1]; identical code base reduces effort, separate boards and cells remove single points of failure.	Common-mode software bug: backup uses different thresholds and delay (Sec. 6.1).
BMP280 (one per Pico)	Pressure/temperature → altitude.	Pressure range 300–1100 hPa covers sea level to well above 3 km (≈701 hPa at 3 km in a standard atmosphere); relative accuracy about ±0.12 hPa (≈ ±1 m) [11]. Barometric altitude is the standard method for apogee and main-deployment decisions.	Port must see static pressure, vented away from fast airflow; lock-out near burnout/transonic transients; absolute offset removed by pad baseline.
MPU6050	Acceleration (and angular rate) for launch/burnout detection and acceleration record.	16-bit ADC, programmable ±2 to ±16 g, 400 kHz I^2C [12], [13]; ±16 g gives 2048 LSB/g, ample resolution.	May saturate during boost: optional ±200 g ADXL375 [14] if simulation shows > 16 g; gyro ±2000°/s limits roll-rate capture.
DS18B20 ×2	Payload temperatures: avionics bay and inner airframe wall.	Digital 1-Wire, factory calibrated, −55 to +125 °C, ±0.5 °C (from −10 to +85 °C), selectable resolution up to 12-bit [15]; many sensors share one wire; 1 Hz sampling matches the conversion time.	Probe thermal lag; mounting bias; cable strain relief. Powered normally (not parasitic).
microSD + SPI module	Non-volatile flight log.	SD cards support SPI mode [16] and are handled by the open FatFs library [17]; ≈26 MB for 90 min at 4.8 kB/s, so even small cards are oversized.	Use a 3.3 V-native socket and a ≤32 GB FAT32 card; log with 512 B blocks and 1 s syncs; test under vibration.
1S Li-ion cell per board (A, B, L)	Power.	Compact, high energy; 3.0–4.2 V maps to the Pico VSYS window [5]; separate cells keep faults isolated (battery safety per [1] 4.7).	Needs protection circuit, fuse, secure holder; verify endurance > 1.5 h.
Arming switch + pull-downs	Safe/armed states.	Hard power gate and gate pull-downs keep outputs low regardless of software state, supporting the safe/armed rules [1].	Accessible from outside the rocket; status code on buzzer/LED.
MOSFET gate drivers + continuity sense	Switch the release device, check the circuit.	Logic-level MOSFETs let 3.3 V GPIO switch higher currents; low-current continuity check proves wiring without activating the release.	Device current and voltage TBD; flyback/EMI protection required.
Buzzer + LED	Self-test and locator.	Immediate pad confirmation of normal operation/arming status [1]; post-landing locator aid.	Not relied on for any flight decision.

Component	Role	Rationale (continued)	Limit / mitigation
JST-XH, labelled harness, PCB/protoboard	Interconnect.	Locking connectors, strain relief and labelled wiring support the vibration and wiring rules [1].	Retention and pull tests.
Official GPS beacon	Location after landing.	Required organiser equipment; kept electrically separate [1].	Installation and cost to confirm.

5. Interfaces and pin allocation (proposed)

Pico	Signal group	Proposed allocation and notes
Logger (L)	I^{2}C0 (GP4 SDA, GP5 SCL), 400 kHz	BMP280 and MPU6050 on one bus; 4.7 kΩ pull-ups (breakouts may already add them). Keep leads short [13].
Logger (L)	SPI0 (GP16–GP19)	microSD, 3.3 V logic; CS pull-up.
Logger (L)	1-Wire via PIO (GP15)	4.7 kΩ pull-up; both DS18B20 on one bus, addressed by ROM code, conversion started for both together.
Logger (L)	ADC0 (GP26)	Cell voltage via resistor divider.
Logger (L)	GP14 buzzer, GP25 LED	Status codes only.
Recovery A/B	I^{2}C0 (GP4/GP5)	Own BMP280 per board; separate sensor, not shared with L.
Recovery A/B	GP10 drogue, GP11 main	To MOSFET gates; external 100 kΩ pull-down on each gate in addition to internal pulls.
Recovery A/B	GP12/GP13, ADC0	Continuity sense inputs and cell voltage.

6. Process flowcharts

Figures 2–5 describe the software processes. Rounded boxes are start/end, rectangles are processes, diamonds are decisions, and amber boxes are fault or degraded paths. Parameters are in Sec. 6.1.

6.1 Decision parameters (initial placeholders pending simulation)

Symbol	Meaning	Starting value / source
a_L, t_L	Launch acceleration threshold and duration (logger and recovery).	e.g. 2.5 g for 50 ms (10 IMU samples); set from motor thrust-to-weight in simulation.
h_L	Altitude rise confirming launch.	e.g. 10 m above baseline within 2 s.
t_{lock}	Apogee lock-out after burnout.	Simulation-derived, e.g. 1–2 s, to avoid pressure disturbances near burnout/transonic flight.
N	Consecutive filtered samples with $\dot{h}_f \leq 0$.	5 samples at 25 Hz (0.2 s).
$T_{apo,max}$	Apogee time-out from launch.	Simulated apogee time + margin.
h_d (A / B)	Drop below peak that confirms apogee on recovery boards.	A: small (e.g. 3 m); B: larger (e.g. 8 m) plus delay Δt_B.
h_{main} (A / B)	Main-parachute deployment altitude (AGL).	Within 150–450 m [1], [7]; B set about 30 m lower than A.
t_{pulse}	Output pulse duration.	From release-device specification TBD.
v_{max}	Baro plausibility limit on $\lvert dh/dt \rvert$.	Simulated peak speed × margin TBD.

7. Robustness and fault handling

Failure mode	Consequence	Mitigation (see flowcharts)
Logger power dip or lockup	Gaps in data or payload record.	Watchdog + scratch-register warm restart; 1 s sync; protected supply.
Barometer noise, spike or port blockage	False apogee or main trigger.	Plausibility gate, IIR filter, multi-condition apogee, lock-out, time-out.
IMU saturation	Bad launch/burnout and acceleration data.	Baro cross-check, ±200 g option [14], flags in record.
SD write stall or removal	Missing records.	Core-1 offload, 24 kB buffer, retry/remount, next-file segment.

Failure mode	Consequence	Mitigation
Recovery A failure	No or late deployment.	Backup B: separate board, sensor, cell, later trigger.
Premature output from noise/boot	Unsafe activation.	Outputs held low by pull-downs, arming switch, launch-inhibit logic.
Thermal probe detaches	Invalid payload data; loose object.	Mechanical retention, strain relief, flag in record.
Connector loosening	Intermittent power/data.	Locking connectors, strain relief, vibration test [1].

The full FMEA follows the organiser template [18]; classical FMECA practice is described in [19].

8. Power and data budgets

Load estimate (logger). Values are datasheet-typical or assumed, and must be replaced by measurements.

Item	mA	Basis
Pico L at 125 MHz, both cores	30	Assumed; measure [6]
MPU6050 (gyro + accel)	3.9	Datasheet typical [12]
BMP280	< 0.1	Datasheet [11]
2× DS18B20 active	2	About 1 mA each [15]
microSD module average	20	Assumed; write peaks higher
Total estimate	**≈56**	Design basis kept at **200 mA**

At 56 mA the 1.5 h minimum [1] needs about 84 mAh. Using the conservative 200 mA at 3.3 V through an assumed 85% efficient converter, the demand is $3.3 \times 0.2 \times 1.5/0.85 \approx 1.16$ Wh, against about 9.6 Wh stored in a nominal 3.7 V, 2.6 Ah cell (≈8× margin). The margin is only a sizing indicator; a measured endurance test of more than 1.5 h is required. Recovery boards use the same method plus the release-device pulse load TBD.

Data budget. 24 B × 200 Hz = 4.8 kB/s; 90 min = 5400 s ⇒ about 26 MB. SPI at 12.5 MHz and 512 B block writes leave large margin, and the 24 kB buffer covers about 5 s of SD latency.

9. Verification plan

Methods follow the usual inspection, analysis, demonstration and test categories [4]; MDR needs the plan, not the results [3].

ID	Method	Planned evidence
V-01	Test	Replay recorded and simulated altitude/acceleration histories through the state machine; check false-apogee and main-trigger margins.
V-02	Test	Logger timing: loop jitter, buffer occupancy and SD latency under load; forced SD errors.
V-03	Demonstration	Watchdog test: force a core stall mid-sequence and confirm state restore and log continuity.
V-04	Demonstration	Recovery A and B each switch safe test loads independently; confirm outputs stay low through power-up, reset and brown-out.
V-05	Test	Power endurance > 1.5 h with realistic logging; connector pull and vibration retention.
V-06	Test	Thermal probes at reference temperature points; data completeness and time alignment.
V-07	Test	Barometer port: pressure-chamber or vented-bay test with representative pressure profile.
V-08	Inspection	Wiring, labels, strain relief, arming-switch access and polarity [1].
V-09	Analysis	Mass, power and cost budget and FMEA update for PDR [18].

10. Decisions needed from the team

1. Approve a three-Pico architecture (logger + two recovery), or choose an alternative recovery controller while keeping the same independence rules.
2. Obtain simulated peak acceleration, speed, apogee time and apogee altitude to fix a_L, t_{lock}, v_{max}, $T_{apo,max}$ and the IMU range.
3. Define the release mechanism with the recovery lead and organiser requirements; replace TBD items with specifications and quotes.
4. Confirm bay dimensions, mounting, battery protection and measured current before procurement.
5. Confirm the official GPS beacon ownership and cost.

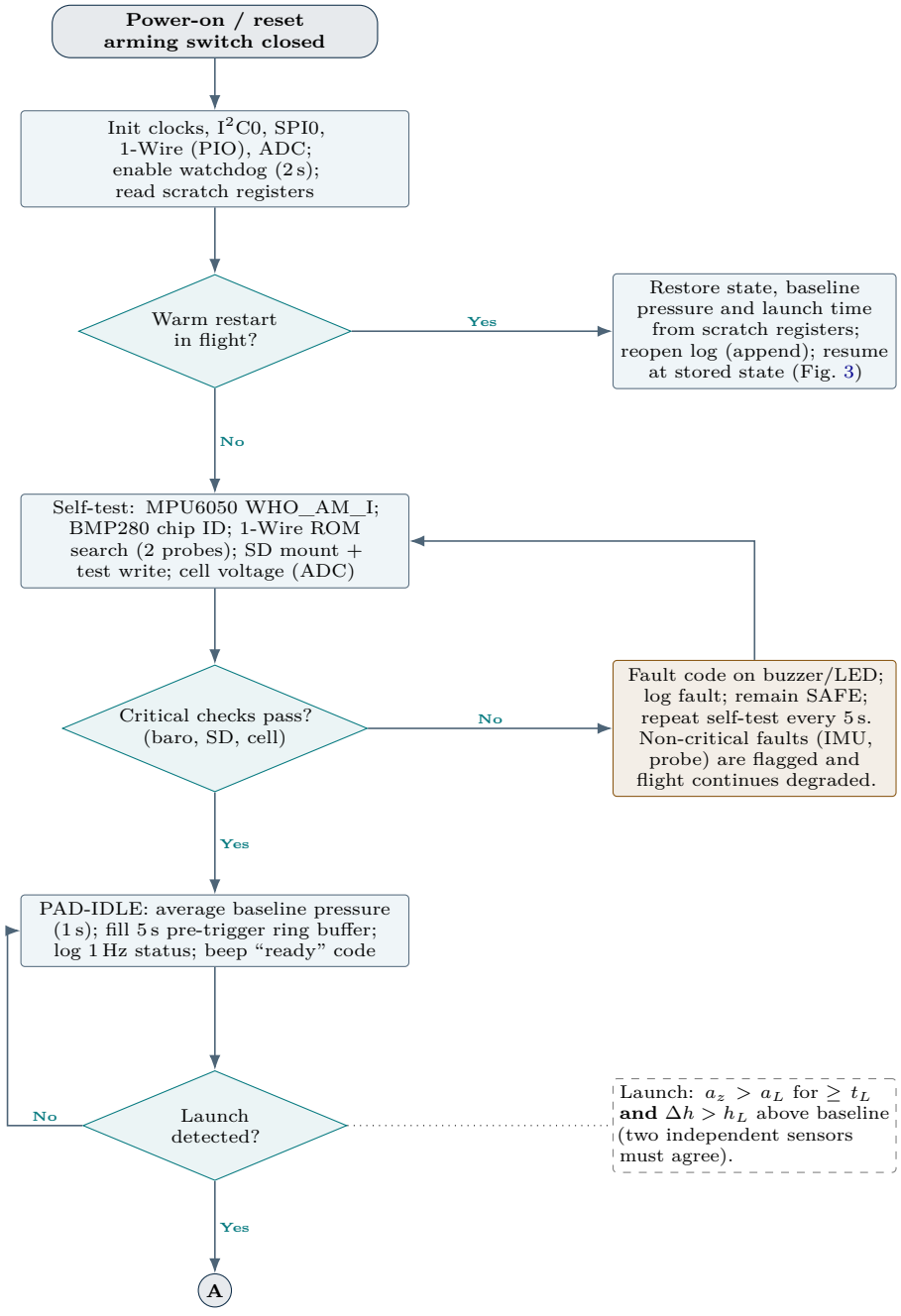


Figure 2: Logger Pico: start-up, warm-restart recovery, self-test and pad-idle until launch detect.

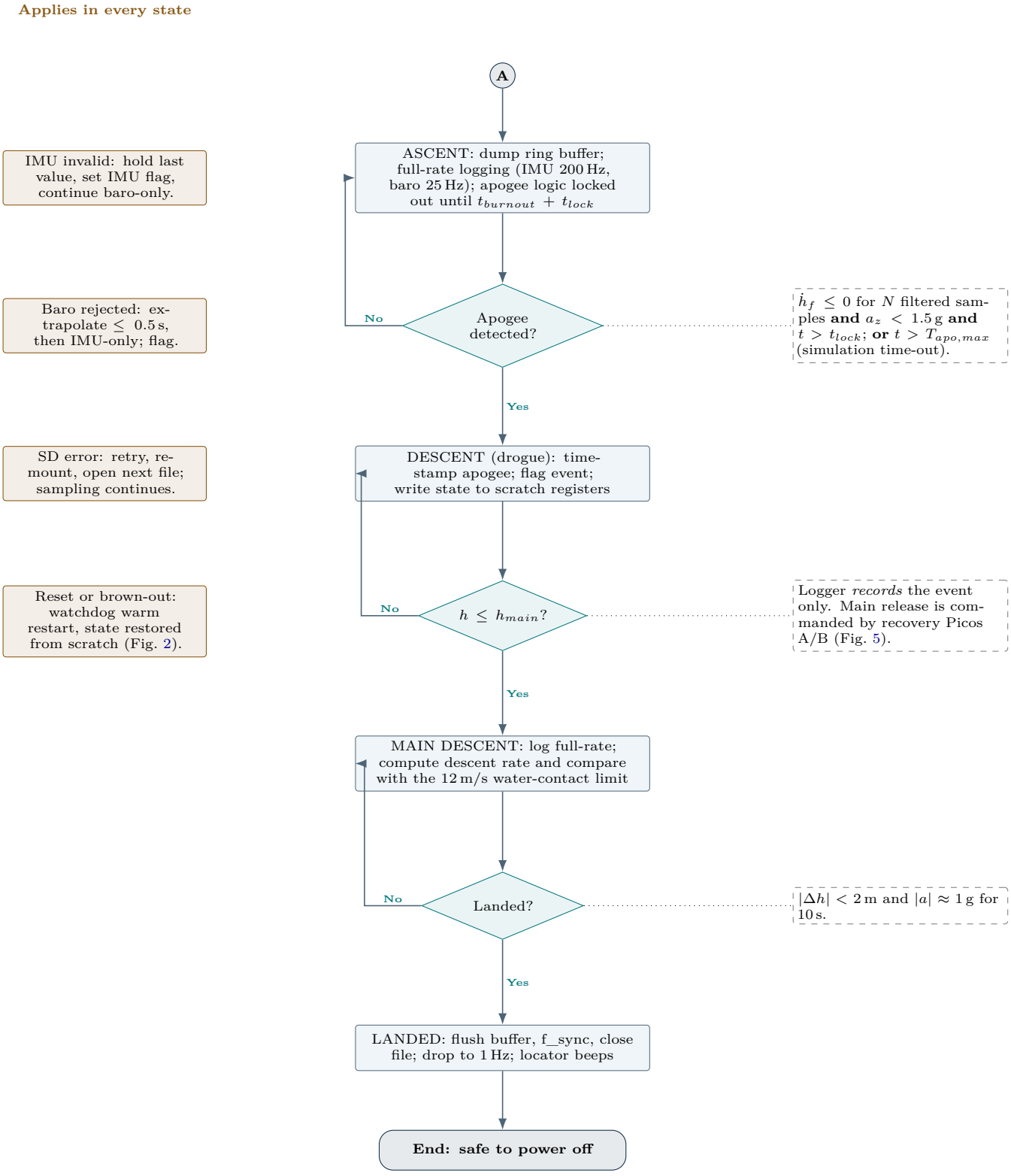


Figure 3: Logger Pico in-flight state machine. Left column: anomaly handling that applies in all states.

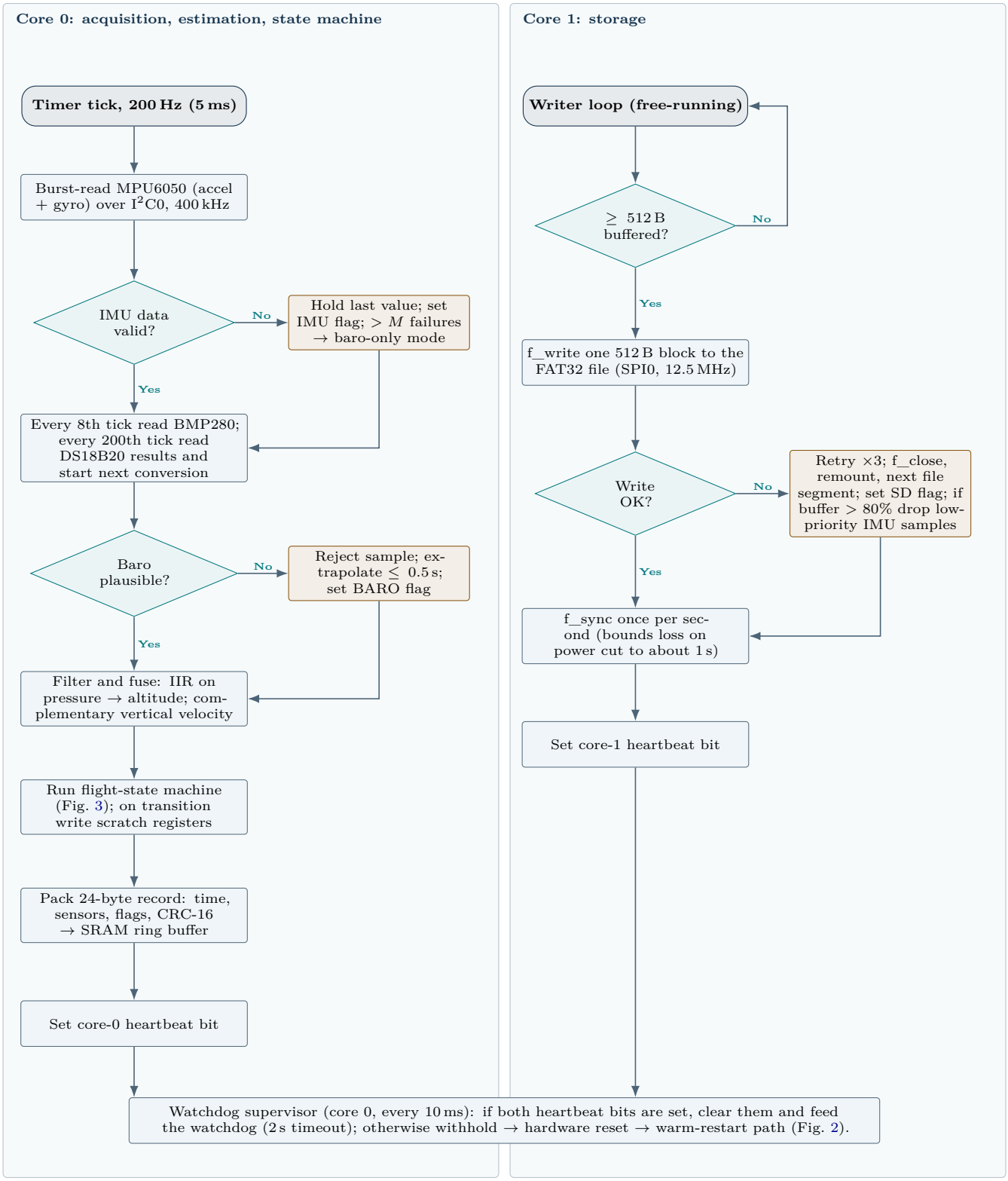


Figure 4: Logger Pico dual-core task flow with fault handling and the cross-core watchdog supervisor. Core 1 absorbs SD latency so core 0 timing is never blocked.

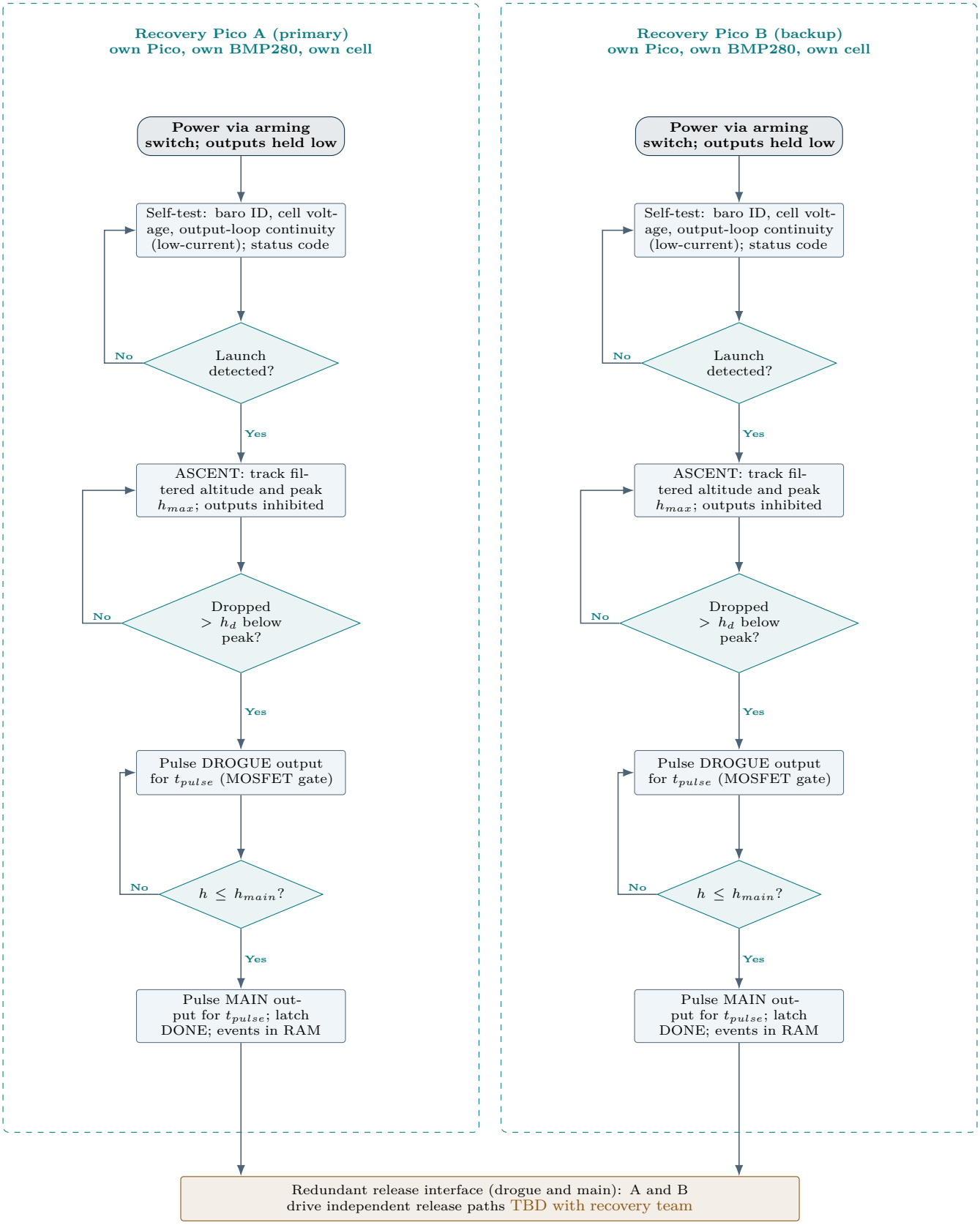


Figure 5: Independent recovery controllers. B uses a later apogee trigger and a lower main altitude than A (Sec. 6.1), so a single common-mode threshold error does not trip both at the same instant.

References

[1] RTC 2027 Organiser, *Appendix 03: Rocket design specifications*, Competition document, `appendix_03.pdf`, 2026.

[2] RTC 2027 Organiser, *Appendix 06: Competition rules explanation, version 2*, Competition document, `appendix_06.pdf`, 2026.

[3] RTC 2027 Organiser, *Appendix 07: Mdr scoring and design maturity matrix*, Competition document, `appendix_07.xlsx`, 2026.

[4] NASA, *NASA systems engineering handbook*, NASA/SP-2016-6105 Rev2, 2016. Accessed: Nov. 4, 2026. [Online]. Available: https://www.nasa.gov/reference/systems-engineering-handbook/

[5] Raspberry Pi Ltd, *Raspberry Pi Pico datasheet: An RP2040-based microcontroller board*, 2024. Accessed: Nov. 4, 2026. [Online]. Available: https://datasheets.raspberrypi.com/pico/pico-datasheet.pdf

[6] Raspberry Pi Ltd, *RP2040 datasheet: A microcontroller by Raspberry Pi*, 2024. Accessed: Nov. 4, 2026. [Online]. Available: https://datasheets.raspberrypi.com/rp2040/rp2040-datasheet.pdf

[7] RTC 2027 Organiser, *Appendix 08: Launch permit inspection checklist*, Competition document, `appendix_08.xlsx`, 2026.

[8] Raspberry Pi Ltd, *Raspberry Pi Pico-series C/C++ SDK*, 2024. Accessed: Nov. 4, 2026. [Online]. Available: https://datasheets.raspberrypi.com/pico/raspberry-pi-pico-c-sdk.pdf

[9] Microchip Technology, *ATmega328P data sheet*, DS40002061, 2020. Accessed: Nov. 4, 2026. [Online]. Available: https://www.microchip.com/en-us/product/ATmega328P

[10] Espressif Systems, *ESP32 series datasheet*, 2024. Accessed: Nov. 4, 2026. [Online]. Available: https://www.espressif.com/en/support/documents/technical-documents

[11] Bosch Sensortec GmbH, *BMP280 digital pressure sensor datasheet*, BST-BMP280-DS001, 2018. Accessed: Nov. 4, 2026. [Online]. Available: https://www.bosch-sensortec.com/products/environmental-sensors/pressure-sensors/bmp280/

[12] InvenSense Inc., *MPU-6000 and MPU-6050 product specification, revision 3.4*, PS-MPU-6000A-00, 2013. Accessed: Nov. 4, 2026. [Online]. Available: https://invensense.tdk.com/products/motion-tracking/6-axis/mpu-6050/

[13] NXP Semiconductors, *I2C-bus specification and user manual, rev. 7.0*, UM10204, 2021. Accessed: Nov. 4, 2026. [Online]. Available: https://www.nxp.com/docs/en/user-guide/UM10204.pdf

[14] Analog Devices, *ADXL375: 3-axis, ±200 g digital MEMS accelerometer*, 2020. Accessed: Nov. 4, 2026. [Online]. Available: https://www.analog.com/en/products/adxl375.html

[15] Maxim Integrated, *DS18B20 programmable resolution 1-wire digital thermometer*, 2019. Accessed: Nov. 4, 2026. [Online]. Available: https://www.analog.com/media/en/technical-documentation/data-sheets/ds18b20.pdf

[16] SD Association, *SD specifications part 1: Physical layer simplified specification*, Use the current published version; SPI-mode command set is defined in this document, 2020. Accessed: Nov. 4, 2026. [Online]. Available: https://www.sdcard.org/downloads/pls/

[17] H. Chan, *FatFs – generic FAT filesystem module*, Online documentation. Accessed: Nov. 4, 2026. [Online]. Available: http://elm-chan.org/fsw/ff/

[18] RTC 2027 Organiser, *Appendix 12: Fmea example*, Competition document, `appendix_12.xlsx`, 2026.

[19] US Department of Defense, *Procedures for performing a failure mode, effects and criticality analysis*, MIL-STD-1629A, 1980.